

80386 Register Organization MPCA Module 5

> **Definition:** The **80386 Register Set** consists of thirty-two 32-bit and 16-bit registers categorized into General-Purpose Registers, Segment Registers, Status and Instruction Registers (EFLAGS & EIP), Control Registers, and Debug/Test Registers.

1. Detailed Technical Explanation

1. General-Purpose Registers (32-bit Extended)

All 16-bit registers of 8086 were extended to 32 bits with the `E` prefix:

31

2. Segment Registers (16-bit)

The 80386 has **6 Segment Selectors** (added FS and GS for additional data segment access):

- `CS`
(General

3. EFLAGS Register (32-bit Extended Flags)

Extends the 8086 flags with critical operating system mode control bits:

31

4. Control Registers (CR0 - CR3)

- **CR0 (Machine Status Word Extended):**

- `PE

2. Quick Recall Flow

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80386

PE/PG